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Informatica® Cloud Application Integration
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Simple React AI Agent with Amazon Bedrock

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Preface

Use *Simple React AI Agent with Amazon Bedrock* to learn how to use Google Search to gather information, perform tasks, and answer user queries. This guide assumes that you have an understanding of the Google API Connector and Amazon Bedrock Connector concepts.

CHAPTER 1

Introduction to Simple React AI Agent with Amazon Bedrock recipe

The Simple React AI Agent with Amazon Bedrock recipe is based on REST and SOAP APIs. The recipe shows you how to build a simple React Agent with a search tool, Google Search, using the Serp API. The React AI agent logic alternates between generating thoughts and performing task-specific actions dynamically.

The Amazon Bedrock Large Language Model (LLM) follows a step-by-step problem-solving approach, utilizing the search tool to gather information and perform tasks to answer user queries. The process orchestrates the whole workflow based on the LLM response.

The process accepts a user query. It passes the query to the LLM with additional system instructions to respond in a format that includes a question, thought, action, action input, observation, final thought, and final answer.

After receiving the response from the LLM, the process checks for the presence of the action and action input fields. If these fields are present, a request for the SearchTool service is formulated. Otherwise, it creates the final answer.

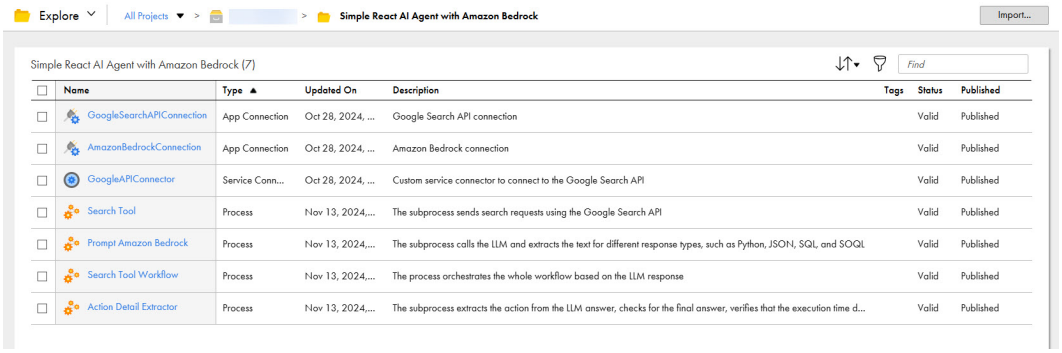
If the final answer is not set, a request is sent to the SearchTool service. The parent process is called with new parameters, such as user query and Agent_Scratchpad.

The process repeats until all the actions from the initial LLM response are processed or the number of repetitions does not exceed the specified limit. When the process execution time approaches 60 seconds, the process provides a final response. If the process ends prematurely, the final response might be inaccurate.

Simple React AI Agent with Amazon Bedrock recipe contents

The Simple React AI Agent with Amazon Bedrock recipe contains app connections, a service connector, and processes.

The following image shows the assets that the Simple React AI Agent with Amazon Bedrock recipe package contains:



Simple React AI Agent with Amazon Bedrock recipe assets

The following table lists the assets that the Simple React AI Agent with Amazon Bedrock recipe package contains:

Asset Name	Asset Type	Description
AmazonBedrockConnection	App connection	Amazon Bedrock connection.
GoogleSearchAPIConnection	App connection	Google Search API connection.
GoogleAPIConnector	Service connector	Custom service connector to connect to the Google Search API.
Search Tool	Process	Subprocess that sends search requests using the Google Search API.
Action Detail Extractor	Process	Subprocess that extracts the action from the LLM answer, checks for the final answer, verifies that the execution time does not exceed 50 seconds, and monitors the number of times the process is called.
Prompt Amazon Bedrock	Process	Subprocess that calls the LLM and extracts the text for different response types, such as Python, JSON, SQL, and SOQL.
Search Tool Workflow	Process	Process that orchestrates the whole workflow based on the LLM response.

CHAPTER 2

Using the Simple React AI Agent with Amazon Bedrock recipe

To use the Simple React AI Agent with Amazon Bedrock recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Publish the GoogleAPIConnector service connector.
3. Configure and publish the Amazon Bedrock connection.
4. Configure and publish the GoogleSearchAPIConnection connection.
5. Configure and publish the processes.
6. Invoke the process.

Step 1. Copy and access the recipe

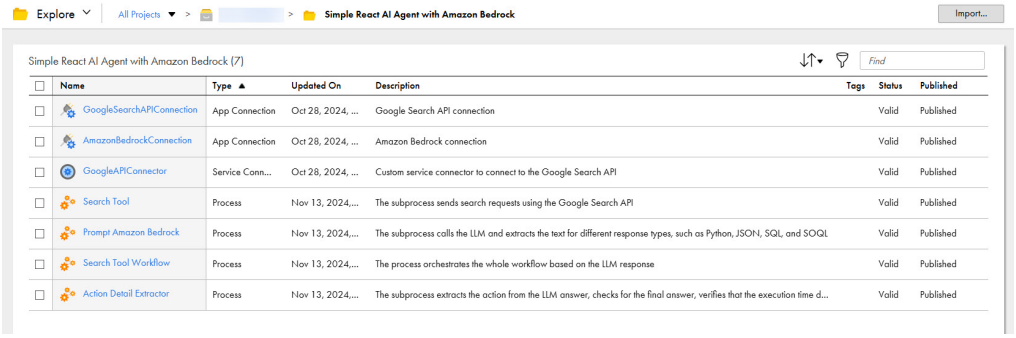
Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Simple React AI Agent with Amazon Bedrock** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.

5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Publish GoogleAPIConnector service connector

You can publish the **GoogleAPIConnector** service connector from the assets page.

To publish the **GoogleAPIConnector** service connector, click **Actions** in the row that contains the service connector and select **Publish**.

Step 3: Configure and publish the Amazon Bedrock connection

Configure the access key, secret key, and region to connect to Amazon Bedrock, and then publish the AmazonBedrockConnection connection.

1. Open the **AmazonBedrockConnection** connection.
2. In the **Connection Properties** section, enter values for the following properties:

Property	Description
AccessKey	The access key to access Amazon Bedrock.
SecretKey	The secret key to access Amazon Bedrock.
Region	The region that hosts Amazon Bedrock.

3. Save and publish the connection.

Step 4. Configure and publish the GoogleSearchAPIConnection connection

Configure the API key in the GoogleSearchAPIConnection connection, and then publish the connection.

1. Open the **GoogleSearchAPIConnection** connection.
2. In the **Connection Properties** section, enter the API key in the **API_KEY** property. The **API_Key** property authenticates Google Search API connection requests.
3. Save and publish the connection.

Note: You can get the API key for the Google search tools from https://serpapi.com/.

Step 5. Configure and publish the processes

Configure the LLM model, loop limit, and prompt request if required and publish the processes.

1. To publish the **Search Tool** process, click **Actions** in the row that contains the process and select **Publish**.
2. To publish the **Action Detail Extractor** process, click **Actions** in the row that contains the process and select **Publish**.
3. Open the **Prompt Amazon Bedrock** process.
4. On the **Temp Fields** tab of the Start step, the **Model_LLM** field is set to **anthropic.claude-3-sonnet-20240229-v1:0** by default. You can optionally edit the model version. For information about changing the model version, see the Amazon Bedrock documentation.
5. Save and publish the process.
6. Open the **Search Tool Workflow** process.
7. On the **Temp Fields** tab of the Start step, the **Loop_Limit** field is set to **5** by default. You can optionally edit the limit. This field defines the number of times the Search Tool subprocess can be called. You can adjust the number of repeat queries in the Search Tool to get new information and better results. However, this will increase the number of tokens used.
8. On the **Assignment** tab of the **Create Prompt to LLM** step, the **Prompt** field is configured to contain the request template for the prompt to LLM. Optionally, you can configure the LLM behavior in the **Prompt_Request** field using the Expression Editor, as shown in the following sample code:

```
For Prompt_Request:
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<messages m:isArray="true">
<role>user</role>
<content m:isArray="true">
<type>text</type>
<text>{$temp.Prompt  }</text>
</content>
</messages>
</root>
```

9. Save and publish the process.

Alternatively, you can also publish the **Search Tool Workflow** process from the assets page. To publish the process, click **Actions** in the row that contains the process and select **Publish**. The subprocesses will be published automatically. You can modify the process as mentioned in steps 7 and 8 before publishing the process if needed.

Step 6. Invoke the process

When you invoke the Search Tool Workflow process, the process orchestrates the whole workflow based on the LLM response.

1. Open the **Search Tool Workflow** process and click **Actions** > **Properties Details** > **Copy Service URL**.
2. Open a text editor and add the input fields and values to the service URL as shown in the following format:
   ```
   <Cloud Application Integration POD URL>/active-bpel/public/rt/<API_name>?Query=<User_Query>
   ```
3. Open a browser and paste the service URL.

You can also invoke the process using the Run Using option, REST or SOAP API endpoints in any API client, such as cURL, Postman, SOAP UI, or through any programming language.

In Application Integration Console, you can verify whether the process execution was successful or faulted.

The following image shows a successful process execution:

